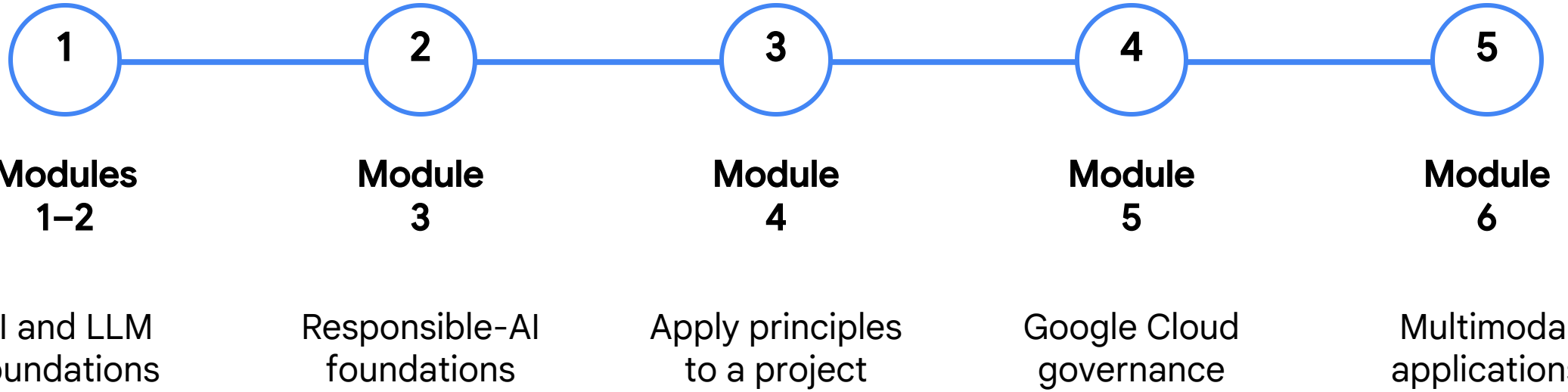
Practical Prompt Design with Vertex AI

04

Applying Responsible AI

Module 4 in the course



Today: a business need becomes a bounded, evaluated, accountable AI service.

01

The business case

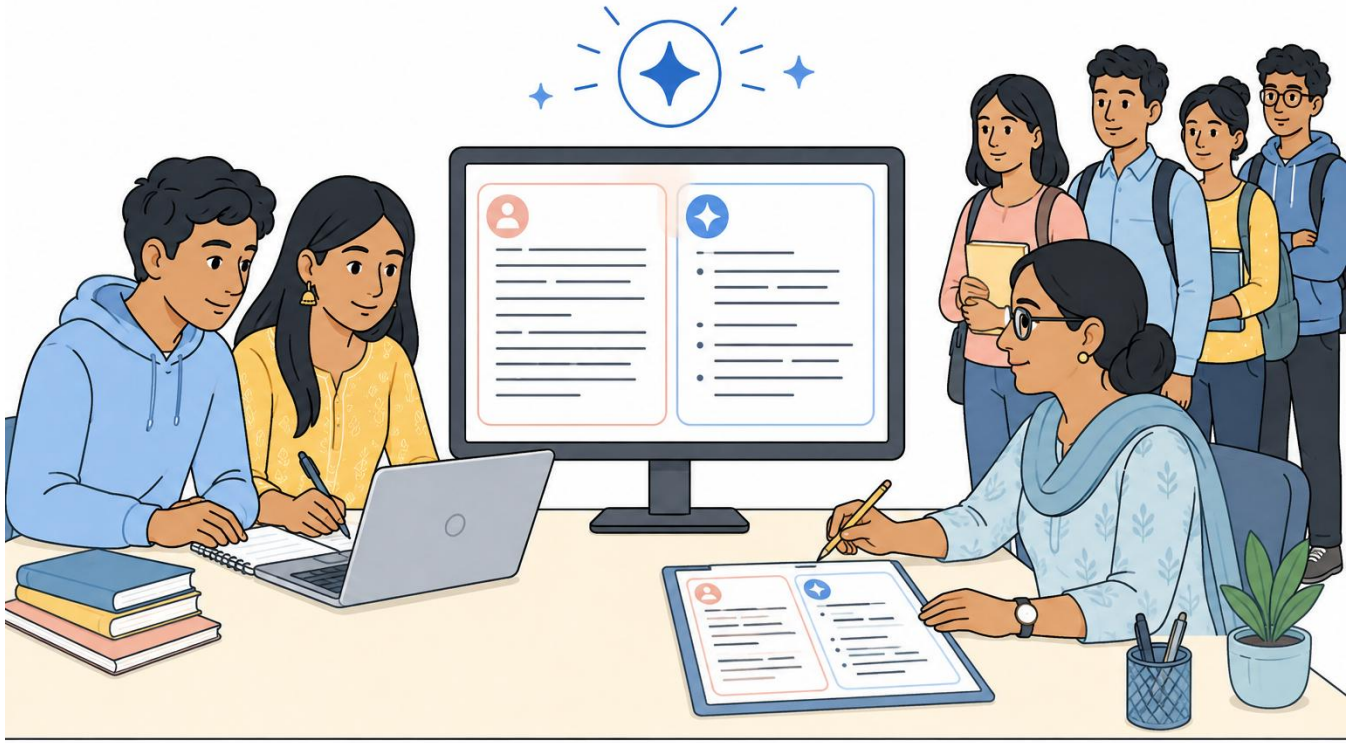
Running case: a campus support assistant

The need

Students wait for the right help. Staff spend time reading and organizing messages.

Proposed role

AI drafts a summary and suggests a route. A trained counselor checks the original request and decides.



Success includes timely, appropriate help and a usable manual route.

A business case connects value with conditions

Need	**Faster first response**	Reduce avoidable waiting while preserving counselor judgment.
Outcome	**Appropriate help**	Track whether the student reaches the right service.
Resources	**People and operating cost**	Budget for review, testing, corrections and model use.
Safeguards	**Bounded pilot**	Keep escalation and manual support available.
Evidence	**A decision to continue**	Expand only when the agreed measures support it.

Technical performance and human outcomes

Measure	What it tells us	Campus example
Efficiency	How quickly work moves	First-response time
Quality	Whether the task succeeds	Counselor agreement with routing
Safety and fairness	Who experiences harmful errors	Critical misses by language group
User experience	Whether people can use the service	Understanding the answer and help route

A low response time cannot establish all four outcomes.

Worked example: review effort and correction effort



Option A

100 initial + 200 corrections
= 300 minutes total

Option B

200 review + 50 corrections
= 250 minutes total

Review takes longer at first, but fewer corrections reduce total effort in this example.

Illustrative numbers only. Count harm and service quality as well as time.

Pilot evidence: faster replies can hide harmful misses

Observed measure	Baseline	Pilot
Average first response	12 minutes	6 minutes
Urgent misses, language A	2 of 100 (2%)	2 of 100 (2%)
Urgent misses, language B	2 of 100 (2%)	8 of 100 (8%)
Uncertain cases using fallback	No AI workflow	3 of 20 cases (15%)

Illustrative data. The pilot improved speed but needs investigation before expansion.

Discussion: should the pilot expand?

“Replies are twice as fast, so the assistant is ready for every student.”

A justified decision

Keep the pilot bounded. Investigate the language pattern, inspect missed cases, and verify that fallback works.

Record the owner, corrective action, retest cases and the conditions for expansion.

Which result would change your decision?

02



Ethical issue spotting

Five questions for an AI project



A useful review asks all five questions

Use the questions before decisions and whenever the use case changes.

The five questions applied to campus support

Question	A specific answer
Purpose	Reduce waiting while preserving qualified judgment.
People	Students, counselors and people needing language or accessibility support.
Failure	A summary omits urgency and delays appropriate help.
Control	A counselor can override routing and use manual support.
Evidence	Critical misses, subgroup results and correction records.

A useful failure statement names the consequence

“A student writes an urgent request in mixed-language text. The summary drops the urgency, and staff route it to a routine queue.”

What the statement gives us

A person, an input condition, a failure in the workflow and a possible consequence.

Now the team can create a test case and decide where a reviewer should intervene.

“The system might be biased” still needs this level of detail.

Fairness example: equivalent need, different wording

Request A

“I cannot access the exam portal.
The deadline is today.
Please help me submit.”

Request B

The same need in another language or with spelling and grammar differences.
The deadline is still today.

Test question: Do both requests receive the same urgency and access to human help?

Compare preserved facts and routing outcomes across supported languages.

Privacy example: the data needed for the task

Too much context

A full student record accompanies a simple timetable question. Unrelated personal details may appear in drafts or logs.

Task-relevant context

The assistant receives the timetable and the question it must answer. Staff confirm permission, limit access and minimize sensitive data.

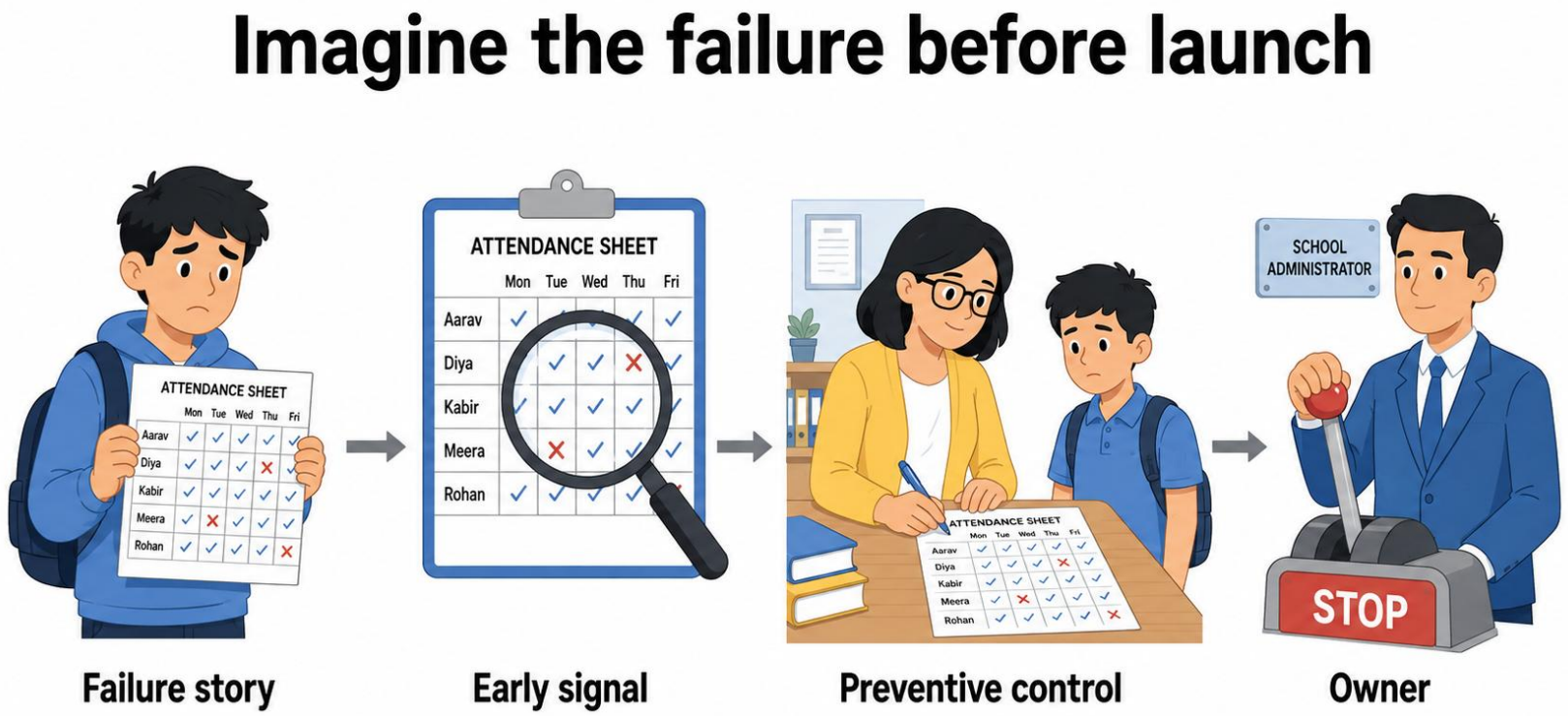
Design decision: remove unrelated fields before they enter prompts, logs or outputs.

Existing access to data does not automatically authorize every new use.

The people analysis includes users missing from testing

Who benefits?	**Instructors save time**	An automated grading proposal reduces marking effort.
Who is missing?	**Students needing accessibility support**	The pilot includes only standard text submissions.
What changes?	**Access and outcome tests**	Include supported submission formats and affected students.
Who acts?	**Course owner and support staff**	Resolve barriers and provide an equivalent manual route.

Premortem: a structured failure rehearsal

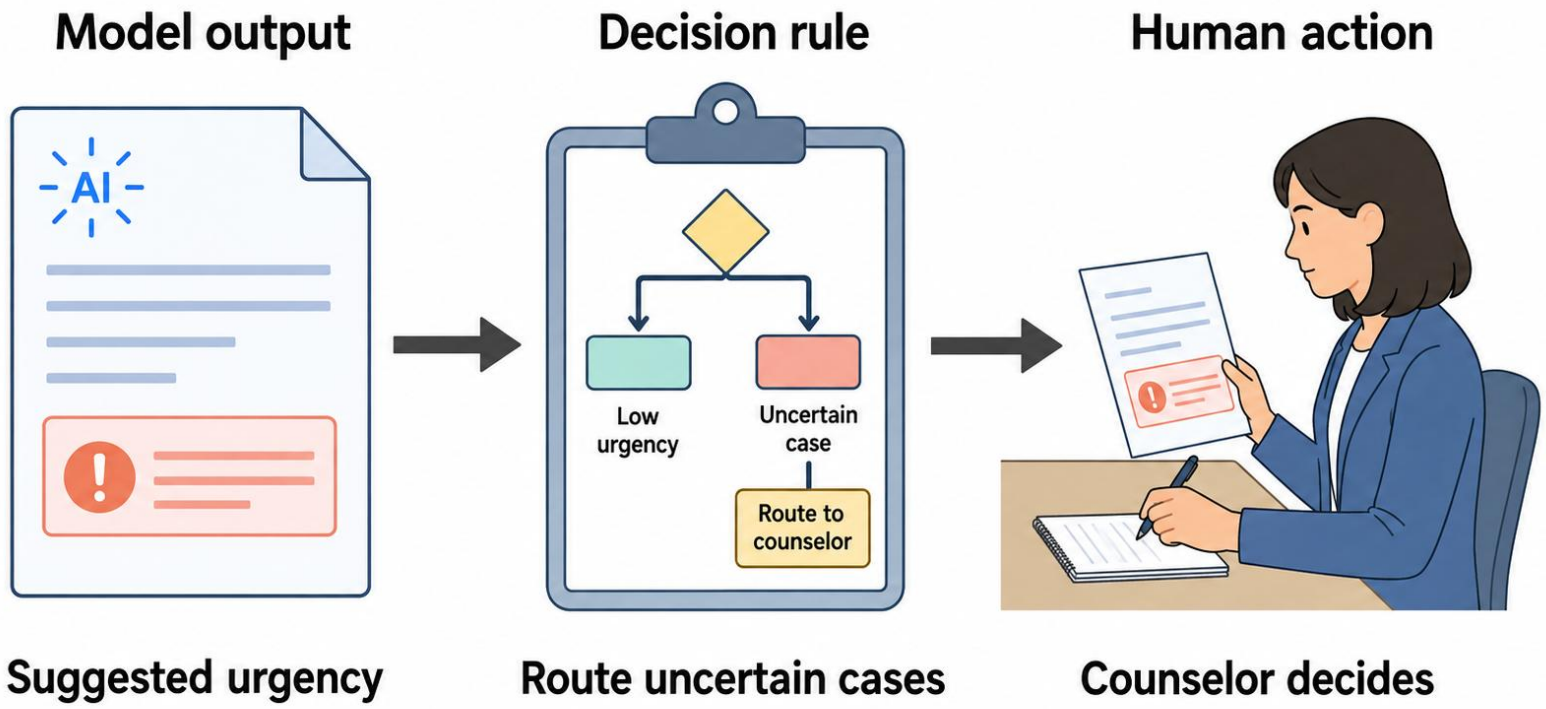


The output is an actionable failure story with a signal, control and owner.

Worked premortem: incorrect attendance records

Part	Example	Why it matters
Failure story	A group repeatedly receives false absence marks.	Attendance records affect students.
Early signal	The group’s false-absence rate rises.	A pattern needs investigation.
Preventive control	Confirm uncertain matches and offer manual attendance.	Errors can be caught before use.
Owner	An authorized staff member reviews and corrects records.	The signal leads to action.

Model output, decision rule and human action



A failure can arise at any of these stages. Test the complete workflow.

Human oversight depends on consequence and uncertainty

Arrangement	How it works	Example
Human-in-the-loop	A person approves before the action.	Counselor approval before urgent routing.
Human-on-the-loop	Automation runs under supervision with intervention authority.	A monitored queue that staff can pause.
Human-out-of-the-loop	No routine prior approval or active supervision of each output.	Low-consequence autocomplete with easy recovery.

In this course, consequential recommendations require stronger review and recourse.

Meaningful review needs more than a reviewer's name

Information

The reviewer sees the original source, uncertainty and known limits.

Capacity and authority

Training, sufficient time and permission to overturn, pause or escalate make review usable.



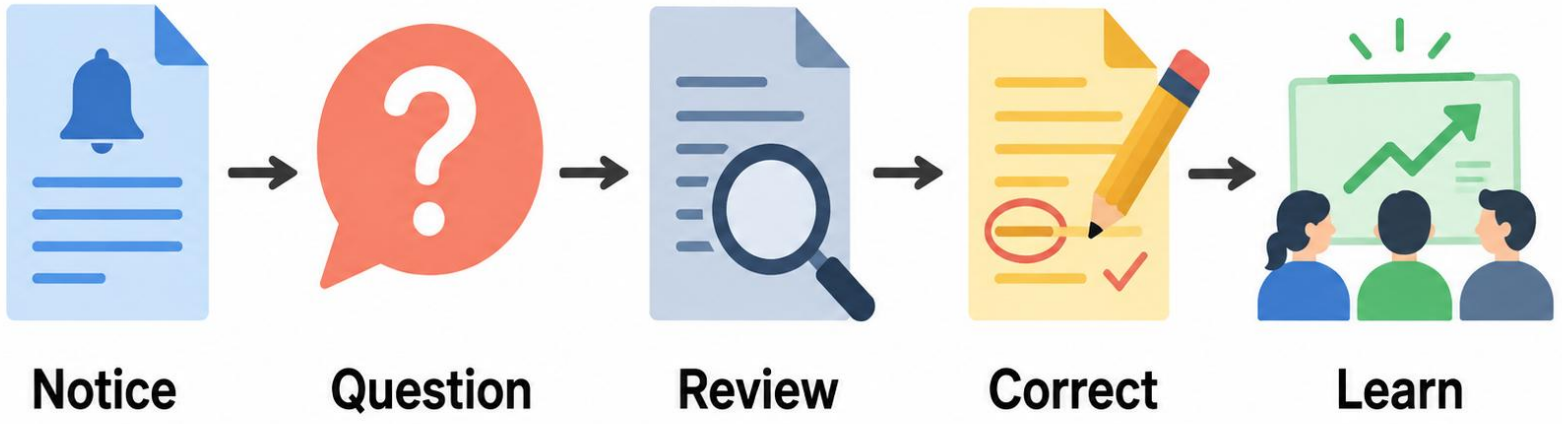
A reviewer who cannot examine or change an output is a nominal safeguard.

Transparency should answer the audience's questions

Audience	Useful information	A practical example
Student	AI involvement, meaning of the outcome and help route.	How to challenge an urgency category.
Reviewer	Original evidence, uncertainty and limitations.	Which source facts the draft may have omitted.
Owner	Tests, incidents and change history.	Whether a revised routing rule reduced misses.

More technical detail does not automatically make an explanation useful.

Recourse includes review and correction



A clear route to challenge and correct an outcome

A notice or complaint form is only one part of the process.

Scholarship scenario: narrowing the AI role

Proposed automation

Rank applicants from historical awards.
Use one generated summary as the decision.
Provide no way to correct missing context.

Responsible redesign

Summarize supplied evidence only.
Trained reviewers check original sources.
Test omissions and subgroup errors.
Applicants can correct factual mistakes.

Boundary: the AI organizes evidence. A qualified reviewer makes and can correct the decision.

The redesign narrows the model's role and adds review and recourse.

Expert assistance in an HR information workflow



The expert needs source access, time and authority to correct the draft.

03

Google’s lessons in practice

Google’s principles evolved with its AI practice



The original seven-principle list is historical. Read the current public principles.

Lessons from Google’s published practice

Review early	**Find issues while design can change**	Review the product and its intended use before commitment.
Include different views	**Improve the questions being asked**	Seek technical, domain and affected-community input.
Change the design	**Act on the review**	Narrow a use, add safeguards or stop when needed.
Keep learning	**Revisit earlier decisions**	Use new evidence to improve the approach.

Historical example: changing an output category

The question Google revisited

Should an image-labeling system assign gender-related labels based on a person’s appearance?

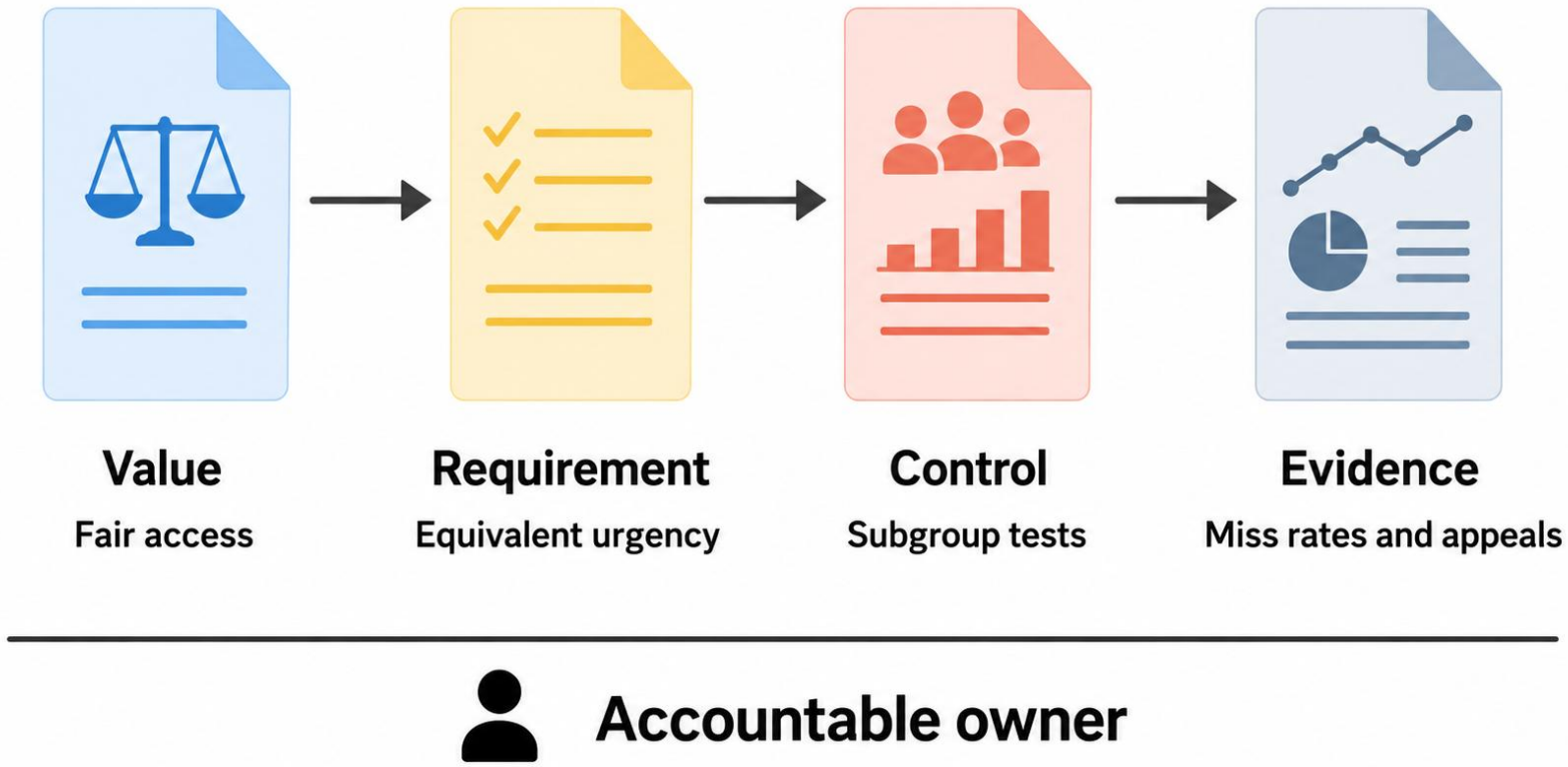
The documented change

Google reported removing gender labels from Cloud Vision API. The permitted output categories changed.

Design lesson: change the permitted output, then verify that the new boundary holds.

Historical example from Google’s published 2020 account.

A principle becomes a requirement and evidence



The owner links the value to daily decisions and checks whether the control works.

Worked chain: fair access to campus support

Value	**Fair access**	Students with equivalent needs can obtain appropriate help.
Requirement	**Language should not change urgency**	Preserve equivalent facts across supported languages.
Control	**Subgroup tests and manual fallback**	Review uncertain or failed cases with a counselor.
Evidence	**Miss rates, appeals and corrections**	Compare outcomes and confirm that fallback operates.
Owner	**Student-support service lead**	Authorize changes when the requirement is not met.

Worked chain: contestability needs records

Requirement and control

Students can challenge an urgency outcome.
A qualified person rechecks the evidence.
That person can correct the outcome and relevant source data.

Evidence to retain

The request for review and its date.
The reviewer’s decision and reason.
Any correction and its completion.
Patterns that need a system change.

Operational test: Can a student use the route, receive review and see a correction completed?

A value, requirement and control still need evidence of operation.

04



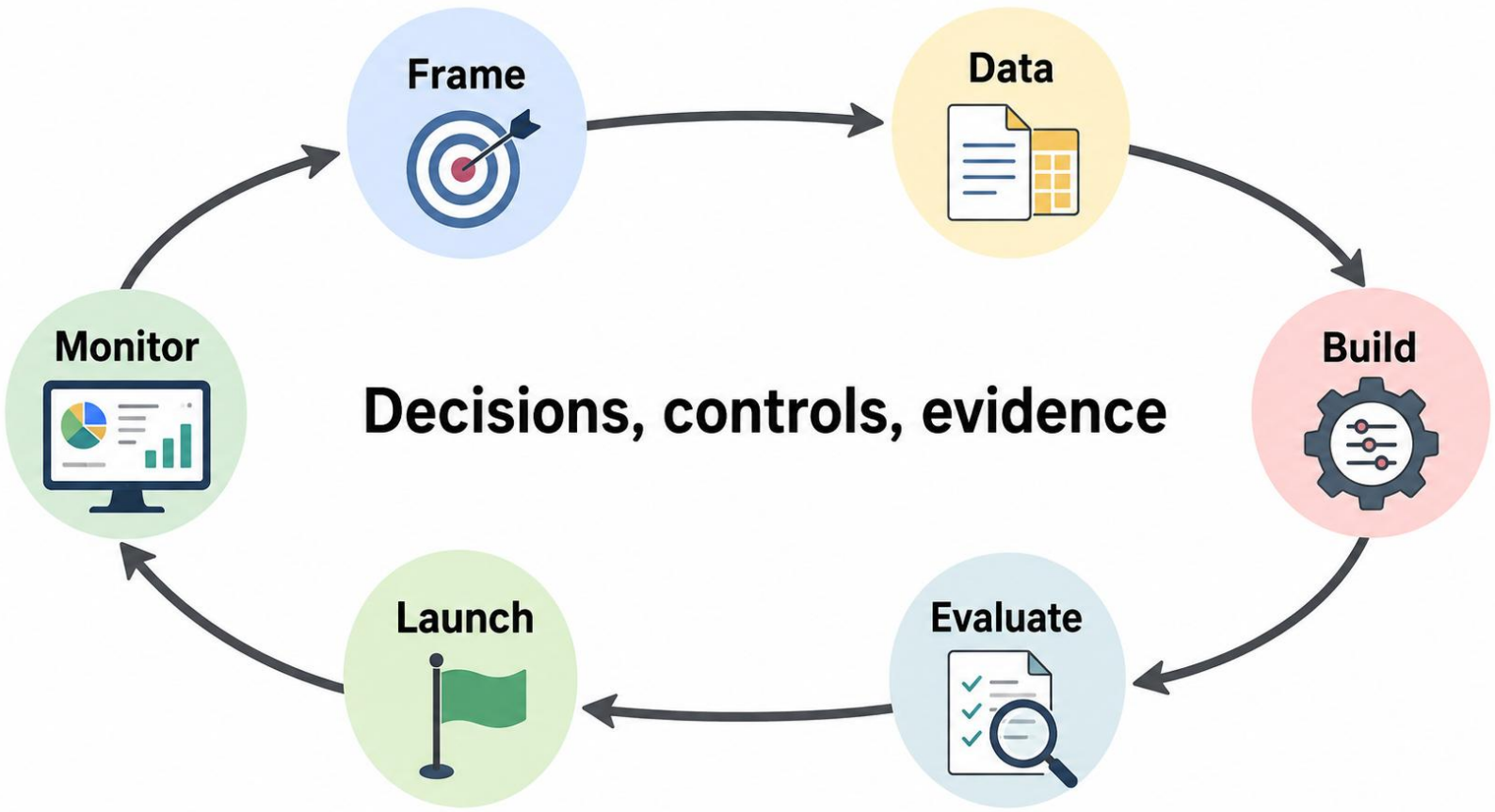
Operationalizing Responsible AI

Lecture 18: ownership, lifecycle decisions and review

An organization needs people, process and resources

Direction	**A defined purpose and values**	Leadership sets boundaries and supports responsible decisions.
People	**Named roles with authority**	Owners, reviewers and affected people contribute.
Process	**Review at meaningful decision points**	Findings can change the task, controls or launch.
Resources	**Tools, training and evidence**	Allocate time for testing, records, feedback and response.

Decisions and evidence across the lifecycle



Revisit earlier stages when monitoring reveals a new need or failure.

Frame and data: the first project records

Frame record

Need: faster access to the right help.
Boundary: AI assists counselor judgment.
Affected people: supported user groups.
Success: useful service and acceptable risk.

Data record

Permission and intended use.
Relevant fields and sensitive-data limits.
Language and accessibility coverage.
Source quality and known gaps.

These records define the claims that later evaluation can support.

Decisions made here determine what later testing must cover.

Build: a bounded role with reviewable evidence

Decision	Control	Evidence
Summarize and suggest a route	Show original source beside the draft.	Source-linked corrections.
Handle uncertain requests	Expose limits and route for qualified review.	Uncertainty and routing records.
Preserve counselor judgment	Allow overrides and manual fallback.	Override outcomes and fallback use.

If confidence scores exist, document and validate what they mean.

Evaluation uses cases that can reveal the failure

Test case	What to inspect	Evidence
Equivalent requests in supported languages	Facts and urgency survive the summary.	Subgroup misses and corrections.
Urgent request with indirect wording	The workflow reaches qualified review.	Critical-miss and escalation results.
Sensitive information in a request	Only needed details reach authorized users.	Disclosure findings and fixes.
Uncertain or unsupported input	Manual fallback actually works.	Observed routing and completion.

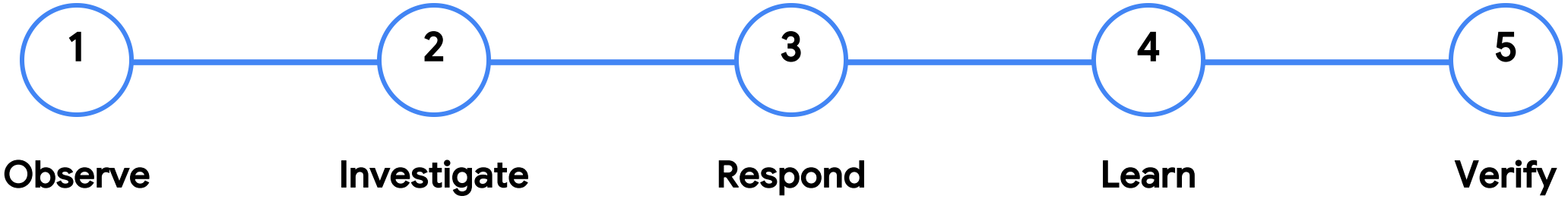
Launch: a pilot with an operating fallback

Scope	**A limited starting group**	Use a clearly defined task and supported inputs.
Notice	**Understandable user information**	Explain AI involvement and the route to help.
Review	**An available qualified reviewer**	Provide time, source access and intervention authority.
Fallback	**A tested manual route**	Verify that people can receive help when AI is uncertain.
Decision	**Conditions for continuing**	Name who can pause, correct or authorize expansion.

Monitoring connects a signal to an action

Signal	Possible concern	Owner's next action
Subgroup misses rise	A pattern affects access or safety.	Inspect cases and pause expansion.
Fallback use drops unexpectedly	A route may be unavailable or bypassed.	Test access and reviewer capacity.
Appeals reveal recurring omissions	The summary drops needed context.	Revise design and add test cases.
Inputs change after launch	Earlier evidence may no longer apply.	Re-evaluate under the new conditions.

A monitoring finding becomes a verified correction



The response owner records what changed and why the service can resume.

A successful pilot has a boundary

What the pilot tested

The original language set.
A limited number of supported tasks.
A particular prompt and routing rule.
The available reviewer workload.

What changed

A new language or input format.
A new consequential use.
An updated model or workflow.
A much larger queue for the same staff.

Decision: identify which earlier assumption changed and select the relevant retest.

Earlier evidence does not automatically support the changed use.

Worked project record: campus support

Stage	Decision or control	Evidence to keep
Frame	Assist counselor judgment.	Purpose and scope record.
Data	Permission, coverage, minimization.	Approved data and gap record.
Build	Source view and uncertain-case review.	Routing and correction records.
Evaluate	Test urgent and subgroup cases.	Miss rates and reviewer findings.
Launch	Bounded pilot and manual fallback.	Launch decision and fallback test.
Monitor	Investigate changes and incidents.	Actions, retests and follow-up.

Trainer demonstration: can a summary preserve the need?

Fictional source message

“The portal rejected my submission. My exam form is due today. I need staff help before the deadline.”

Compare two prepared drafts

Draft A: “The student has a portal question.”
Draft B: “The portal rejected an exam-form submission due today. Staff help is requested.”

Observe omissions, inspect the source and decide what still needs human review.

Group activity: prepare a review decision

A short review-board exercise

- Choose attendance, grading or customer support.
- Complete the five issue-spotting questions.
- Write one requirement, control, owner and evidence source.
- Decide: revise the design, continue a bounded pilot, or stop.

Use a specific failure scenario to explain your decision.

Google Skills learning activity

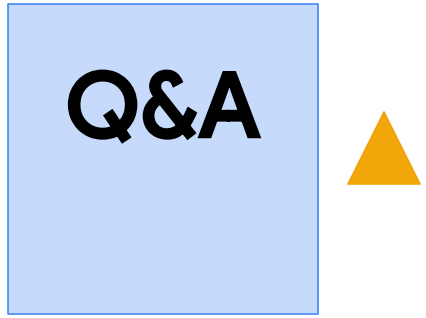
Responsible AI: Applying AI Principles with Google Cloud

- Complete the course's available lessons and assessment.
- Relate one lesson to the campus support project.
- Identify one control and the evidence needed to judge it.

This is a conceptual course activity, not a timed Google Cloud lab.

Module review

Business case	**Useful outcomes with safeguards**	Value includes service quality and correction effort.
Issue spotting	**Specific people and failures**	Purpose, people, failure, control and evidence.
Principles in practice	**Values become decisions**	Requirements, controls, records and accountable owners.
Operation	**Evidence throughout the lifecycle**	Review changes and verify that corrective action works.



Knowledge check

Three questions, followed by explanations

Question 1: A pilot is faster, but one language has more urgent misses. What is the best next step?

A. Expand because average speed improved.

B. Investigate subgroup failures and verify fallback before expansion.

C. Remove language-level metrics from the report.

D. Use a larger model and skip the retest.

Answer 1

B. Investigate subgroup failures and verify fallback

Speed is one outcome. The team needs evidence that urgent requests receive appropriate handling across the supported population.

Question 2: A reviewer can see the draft but cannot read the source or change the result. What is missing?

A. A more prominent reviewer name.

B. A higher number of model parameters.

C. More draft text on the screen.

D. Information and authority for meaningful oversight.

Answer 2

D. Information and authority for meaningful oversight

The reviewer needs source context, uncertainty, time, training and the ability to overturn or escalate the result.

Question 3: A team promises fair access and adds manual fallback, but records no outcomes. What is the main gap?

A. Evidence showing whether the requirement and control work.

B. A longer public statement about its values.

C. More colors in the interface.

D. Automatic approval of every future use.

Answer 3

A. Evidence that the requirement and control work

Values guide requirements. Controls put them into practice. Evidence and accountable owners support the next decision.

References: principles and their development

Google AI Principles

https://ai.google/principles/

AI at Google: our principles (2018, updated 2025)

https://blog.google/innovation-and-ai/products/ai-principles/

Taking care of business with Responsible AI (2020)

https://cloud.google.com/blog/products/ai-machine-learning/taking-care-of-business-with-responsible-ai

Responsible AI Progress Report 2026

https://ai.google/static/documents/ai-responsibility-update-2026.pdf

Documentation checked: 5 September 2026

References: design, evaluation and safeguards

Design a responsible approach

https://ai.google.dev/responsible/docs/design

Evaluate model and system for safety

https://ai.google.dev/responsible/docs/evaluation

Safeguard your models

https://ai.google.dev/responsible/docs/safeguards

Documentation checked: 5 September 2026

References: human interaction and learning

People + AI Guidebook

https://pair.withgoogle.com/guidebook/

Responsible AI: Applying AI Principles with Google Cloud

https://www.skills.google/course_templates/388

Responsible AI at Google Cloud

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Documentation checked: 5 September 2026